

Jiyang Lee

📍 Gwanak-gu, Seoul, South Korea ✉ jiyang.lee@snu.ac.kr

Research Interest

Robotic Learning, Embodied AI, Vision-Language-Action Models

Education

Seoul National University, Seoul, Korea *Mar 2024 – Present*
Ph.D Student in Civil & Environment Engineering
Advisor: Prof. Seong-Woo Kim

University of Waterloo, Waterloo, Canada *Jan 2026 – Jul 2026*
International Visiting Graduate Student
Advisor: Prof. Roya Firoozi

Sogang University, Seoul, Korea *Mar 2015 – Feb 2017*
Master of Science in Mechanical Engineering
Advisor: Prof. Cheol-soo Lee
Graduate Research Assistant, CAD/CAM Lab (03/2015–02/2017)

- Analyzed PET dispensing system and built a robotic dispensing system.
- Designed multi-axis manipulators and customized a SCARA robot.
- Published two journals and attended three conferences.

Sogang University, Seoul, Korea *Mar 2011 – Feb 2015*
Bachelor of Science in Mechanical Engineering
Undergraduate Research Assistant, CAD/CAM Lab (12/2013–02/2015)

- Researched Six-Axis Robot Manipulator mechanisms.
- Awarded Academic Excellence Scholarships.
- Admitted with highest distinction on College Entrance Exam.

Professional Experience

LG Display Co., Ltd., Paju, Korea *Dec 2020 – Feb 2024*
Research Engineer, Vision Algorithm Team

- Developed Few-Shot Learning Classification Model for OLED defect images.
- Built AI-based Anomaly Detection Automatic Alarm System.
- Accelerated legacy rule-based anomaly system with CUDA, improving speed by 7 times.

Korea Institute of Science and Technology (KIST), Seoul, Korea *Oct 2018 – Feb 2020*
Robotics Researcher

- Designed and assembled a quadruped robot.
- Developed panoramic view system for Firefighting Armored Robot.
- Conducted UWB-based robot swarm localization experiments.

CSCAM Co., Ltd., Gwangju, Korea *Jan 2017 – Sep 2018*
CNC Department

- Devised Smartphone Frame Vision Inspection machine.
- Built XRF Inspection Machine for Samsung Electro-Mechanics.
- Designed loading and buffer machines for OLED manufacturing lines.

Publications

Journals

- **Lee, J.Y.** and Lee, C.S. (2018). *Path planning for SCARA robot based on marker detection using feature extraction and labelling*. International Journal of Computer Integrated Manufacturing, 31(8), 769-776.

- **Lee, J.Y.**, Lee, D.G., and Lee, C.S. (2017). *Development of Robotic Dispensing System for Radiopharmaceuticals using SCARA Robot*. Journal of the Korean Society of Manufacturing Technology Engineers, 26(5), 441-449. [Best Paper Award]

Conferences

- Chan Kim, Keonwoo Kim, Mintaek Oh, Hanbi Baek, **Jiyang Lee**, et al., "E2Map: Experience-and-Emotion Map for Self-Reflective Robot Navigation with Language Models," IEEE ICRA, Atlanta, USA, May 2025.
- Kim, M., **Lee, J.Y.**, et al., "A Robust Position Estimation Algorithm under Unusual Large Range Errors," IEEE ROMAN, New Delhi, India, 2019.
- **Lee, J.Y.** and Lee, C.S., "Path planning for SCARA robot based on marker detection using feature extraction, labeling, and inverse perspective transform" 26th International Conference on Flexible Automation and Intelligent Manufacturing, Seoul, 2016.
- Lee, C.S., **Lee, J.Y.**, et al., "Teleoperation of SCARA Robot for Radiopharmaceutical Product Dispensing," Korean Society of Manufacturing Technology Engineers, Fall Conference, 2016.
- **Lee, J.Y.** and Lee, C.S., "SCARA Robot 2D Path Planning using Markers," Winter Conference of Society for Computational Design and Engineering, 2016.

Research Experience

- Graduate Teaching Assistant, MEE6413: Intelligent Actuator (Aug. 2017 – Dec. 2017)
- Graduate Teaching Assistant, MEE4004: 3D CAD & Practice (Mar. 2015 – Feb. 2017)
- Grader, MEE2026: 2D-CAD (Mar. 2015 – Feb. 2017)

Awards

- Best Paper Award, Korean Society of Manufacturing Technology Engineers, 2017
- Robot Membership Award, Samsung Techwin, 2014
- Excellence Award, TI Innovation Challenge: Korean MCU Design Contest, 2013

Additional Information

- **Programming Skills:** C++, Python, OpenCV, PyTorch, CUDA
- **Mechanical Design:** Autodesk AutoCAD, Inventor, SolidWorks